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In UK visiting Ken Carslaw at the University of Leeds

What role do clouds play in Arctic climate change?

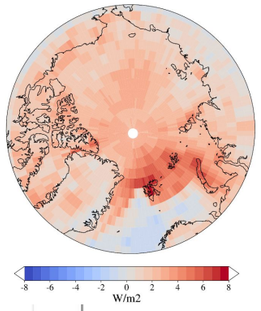
STEP CHANGE

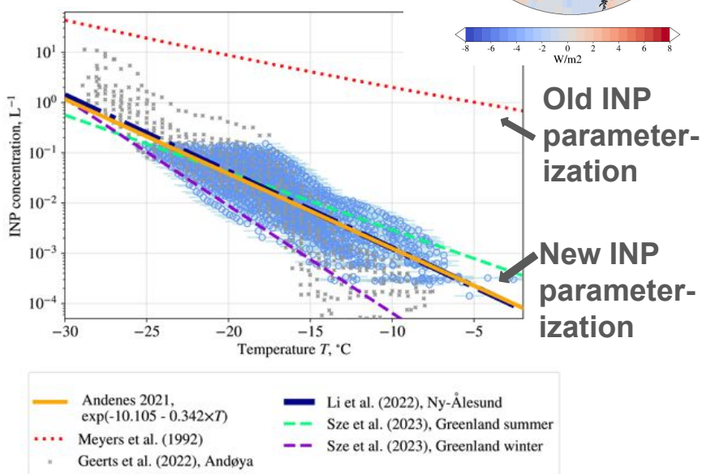
UNIVERSITETET I OSLO

Big question 1: How to model Arctic cloud phase?

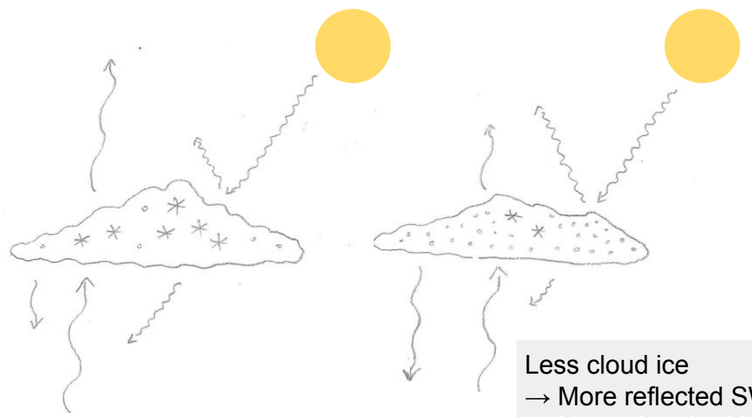
The representation of ice-nucleating particles (INP) in climate models matters for present-day Arctic cloud phase - how much does it matter for cloud feedback?

Net difference in present-day in cloud radiative effect change at surface





Cloud phase matters for climate…



Less cloud ice
→ More reflected SW
→ More absorbed LW

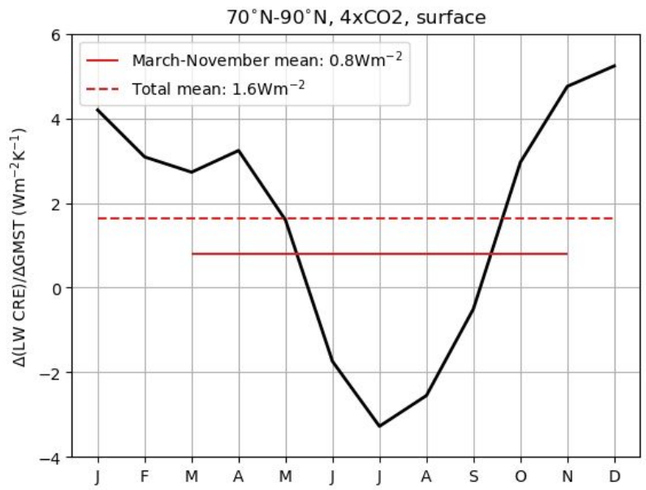
… but how will Arctic cloud phase change with warming, and how will this affect climate?

If I had endless computational resources…
(I think) I would create a perturbed parameter ensemble for cloud phase related a climate model parameters to investigate how Arctic cloud feedback is sensitive to realistic values of INPs, secondary ice production, and other cloud phase controlling mechanisms. Would you?? I'd love to discuss:)

Big question 2: How to calculate Arctic cloud feedbacks?

Cloud feedback calculation from satellite simulators are very useful to decompose cloud feedbacks in climate models using cloud radiative kernels!

However, they do not capture cloud changes during polar night – the LW cloud feedback may be underestimated. I want to look into cloud radiative kernels that are more suited for studying the Arctic!



The seasonal cycle of LW CRE change from piControl to 4xCO2 in NorESM2.3, normalized by global surface mean temperature change.

LW CRE = LW(all-sky) - LW(clear-sky), downwelling direction positive

Anything you want to discuss? Please contact me at astridbg@uio.no